Assignment for Week 9/10

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Package Loading

```
library(ggplot2)
library(dplyr, warn.conflicts = F)
library(readr)
```

Assignment

```
URLchunkA <- "https://raw.githubusercontent.com/washingtonpost/"
URLchunkB <- "data-police-shootings/master/fatal-police-shootings-data.csv"
shootings <- read_csv(paste0(URLchunkA, URLchunkB))
```

```
## Parsed with column specification:
## cols(
##   id = col_integer(),
##   name = col_character(),
##   date = col_date(format = ""),
##   manner_of_death = col_character(),
##   armed = col_character(),
##   age = col_integer(),
##   gender = col_character(),
##   race = col_character(),
##   city = col_character(),
##   state = col_character(),
##   signs_of_mental_illness = col_character(),
##   threat_level = col_character(),
##   flee = col_character(),
##   body_camera = col_character()
## )
```

1) This question asks us to use ggplot to display the timing of police shootings data above.

a) First, use the `lubridate` package to round dates to the nearest month. Use the `round_date` function.

b) Display these data with a histogram with 10 bins, 20 bins, 30 and 35 bins. Do the same with a density plot with varying adjustment equal to .1, .2, and .8. Describe what is happening.

c) Produce a table which displays the count of white people that were shot and the presence of a body camera.

d) What would the expected counts be if these two variables (being white and body camera presence) were independent?

e) What is the difference between the observed counts and the expected counts?

f) Square the answers from e), divide by the expected count, and take the sum. This is called the chi-square statistical test of independence. It is distributed chi-square with `2*2 -1 =3` degrees of freedom.

g) Use the cdf function for the chi-square to describe the probability of observing the result from f) were these two variables independent.

h) Replicate the above steps with the chisq.test command.

2) Use the `lakers` dataset from the lubridate package to determine the following:

a) Which top 40 Laker's performance (in terms of made free throw shots) was most affected by being home or away?

b) Which top 40 Laker's performance improved most (in terms of made free throw shots) when the lakers were losing?

c) ADVANCED/Extra Credit: Plot the distribution of time between shots over the course of games.

3) Come up with an exam question on each of the following concepts, these can be multiple choice, word problems, or True/False. Be sure to provide an answer:

a) The inclusion/exclusion principle.
b) Confidence intervals when estimating the variance.
c) The difference between independence and mutually exclusive.
d) The determining whether a function is a pdf.
e) Markov's inequality.
f) Law of Total Probability.
g) Law of Iterated Expectations.
h) Calculating a variance.
i) The difference between μ and $\bar{X}$
j) Jensen's inequality.
k) Joint and marginal distributions
l) Tranformations of random variables.